Q1.

Suppose you have the sentence:

> "I love machine learning today"

Using a **Skip-gram** model with window size = 1:

1. What are the input-target pairs generated?
2. Which word is the input when predicting "I"?
3. Which word is the input when predicting "machine"?

Q2.

Why do we use Word2Vec instead of One-Hot Encoding?

Explain in terms of:

- Dimensionality
- Similarity between words
- Memory usage

Q3.

What is the difference between:

- CBOW
- Skip-gram

Which one predicts context from a target word?